DiffResist: Physics-Constrained Diffusion for Photoresist Modeling

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Key Results

DiffResist reformulates 3D photoresist profile prediction as a physics-constrained 2D diffusion problem, achieving **state-of-the-art accuracy** (EPE-mean **2.56 nm**) with $>10\times$ **faster** inference than 3D diffusion baselines.

Motivation

Accurate 3D photoresist simulation is essential for optical lithography at advanced nodes, but existing learning-based methods face a key trade-off:

- **Direct mapping** (3D learning): demands prohibitive compute for the high-dimensional output.
- **Sequential modeling**: efficient but suffers from **error accumulation** along depth.
- **DiffResist (ours)**: a physics-constrained 2D diffusion that corrects predictions and suppresses error propagation.

Unlike open-ended video generation, 3D resist simulation benefits from **well-studied exposure physics** that act as strong priors to regularize learning.

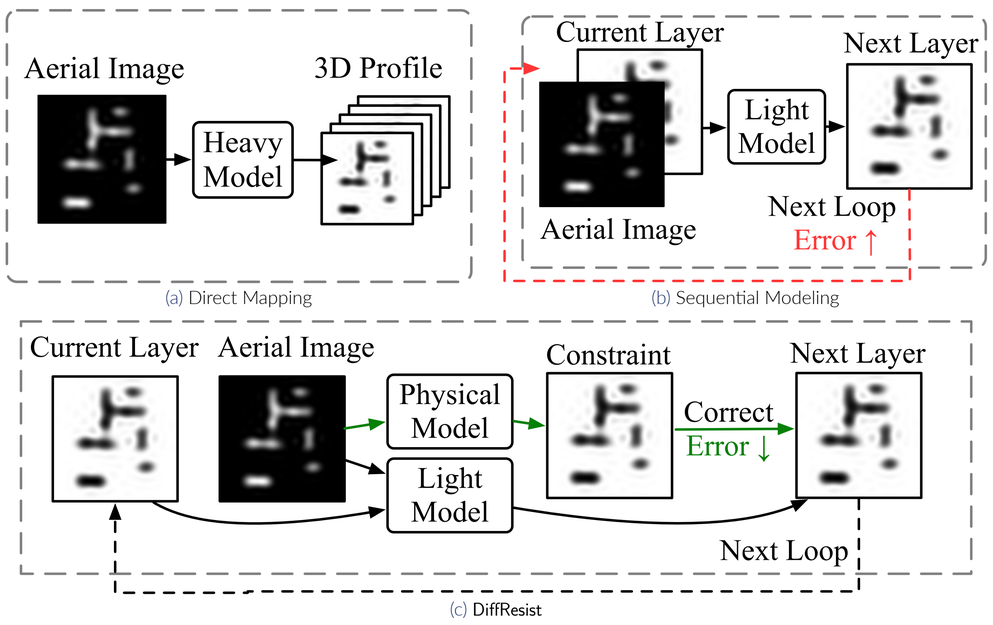

Figure 1. Modeling paradigms. (a) Direct mapping is costly. (b) Sequential modeling accumulates errors. (c) **DiffResist** injects physical constraints to correct sequential predictions.

Resist Exposure Physics

Under the Dill model [2], intensity $\mathbf{I}(h,t)$ and inhibitor concentration $\mathbf{M}(h,t) \in [0,1]$ satisfy coupled PDEs:

$$\frac{\partial \mathbf{I}}{\partial h} = -\mathbf{I}[A\mathbf{M} + B], \qquad \frac{\partial \mathbf{M}}{\partial t} = -\mathbf{I}\mathbf{M}C, \tag{1}$$

where A, B, C are known constants. The aerial image $\mathbf{R}$ provides a **closed-form boundary condition** at the resist surface $h=0$:

$$\mathbf{M}_0(\mathbf{R}) = \exp(-\mathbf{R}\,C\,T). \tag{2}$$

Pipeline Overview

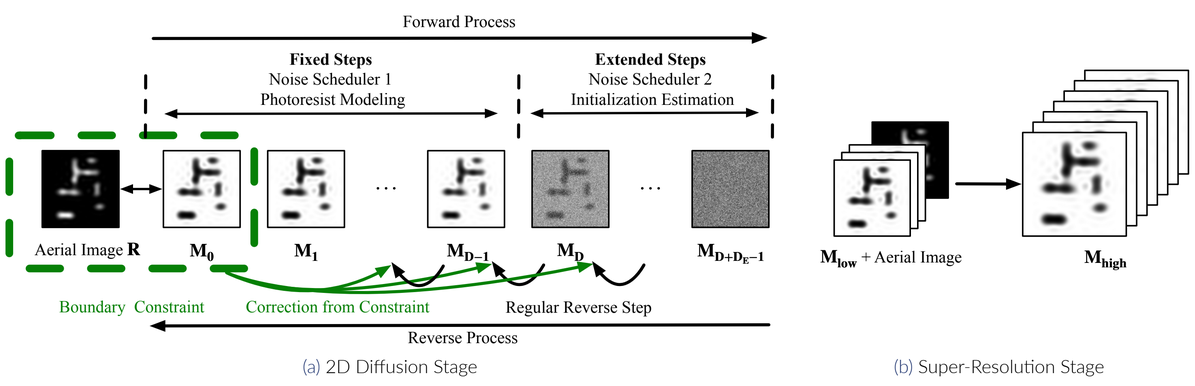

Figure 2. **DiffResist**: (a) Physics-constrained 2D diffusion predicts a low-res 3D inhibitor volume. (b) Super-resolution upsamples to target resolution.

Physics-Constrained 2D Diffusion

Forward Process (Two Stages)

We represent the 3D volume as D slices $\mathbf{M}_d \in [0,1]^{H\times W}$. **Stage 1** — Physical layers ($0 < d < D$): deterministic from data, $q(\mathbf{M}_d|\mathbf{M}_0) = \delta(\mathbf{M}_d - \mathbf{M}_d^{\text{data}})$. **Stage 2** — Warm-up ($D \le d < D{+}D_E$): noised to connect to $\mathcal{N}(\mathbf{0},\mathbf{I})$.

Reverse Process with Boundary Condition

Starting from $\mathbf{M}_{D+D_E-1} \sim \mathcal{N}(\mathbf{0},\mathbf{I})$, we inject the **exact boundary condition** (2):

$$\mathbf{M}_{d-1} = \underbrace{\sqrt{\bar\alpha_{d-1}}\,\mathbf{M}_0(\mathbf{R})}_{\text{Corrected top layer}} + \underbrace{\frac{(1-\bar\alpha_{d-1})\sqrt{\alpha_d}}{\sqrt{1-\bar\alpha_d}} \cdot \boldsymbol{\epsilon}_\theta(\mathbf{M}_d, d, \mathbf{R})}_{\text{Directional correction}} + \underbrace{\sigma_d \mathbf{z}}_{\text{Noise}}, \tag{3}$$

where $\boldsymbol{\epsilon}_\theta$ is a U-Net conditioned on $\mathbf{R}$.

Two-Stage Noise Schedule

$\sigma_d^2 = 0$ for physical layers (no error accumulation); standard DDPM [3] variance for warm-up steps. Training uses simplified noise-prediction loss; for $0<d<D$ the target noise is **deterministic** (known from data).

Lightweight Super-Resolution

Down-sampling $(75,147,147) \to (15,49,49)$ by $(\lambda_H,\lambda_W,\lambda_D) = (3,3,5)$. A residual network (**0.48M** params) recovers full resolution: $\mathbf{M}_{\text{high}} = g(\mathbf{M}_{\text{low}},\mathbf{R}) + \text{UP}(\mathbf{M}_{\text{low}})$, where g is a small CNN and UP is bilinear interpolation. Final printed pattern is obtained via the bulk development model.

Quantitative Results

Evaluated on LithoBench [5] (16,472 tiles) with TorchResist [4] 2000-step ground truth (single Nvidia 3090).

Baselines: 3D-DDPM, 3D-DDIM (direct 3D diffusion); SVG [1], SVG-RNN (sequential).

Table 1. Results on LithoBench. PD: Pixel Difference (%). EPE: nm.

Method	Params	FLOPs	Train (hrs)	MSE	PD	EPE-max	EPE-mean
3D-DDPM	25.61M	15.53T	146.2	4.14×10^{-2}	19.10	40.47	10.17
3D-DDIM	25.61M	15.53T	146.2	3.79×10^{-2}	18.54	39.52	9.86
SVG	8.73M	99.91G	7.3	1.41×10^{-3}	2.18	19.14	7.73
SVG-RNN	7.15M	99.82G	7.5	7.39×10^{-4}	1.02	17.06	4.62
DiffResist	6.71M	69.57G	7.0	$\mathbf{4.17\times10^{-4}}$	**0.85**	**13.64**	**2.56**

- Direct mapping (3D-DDPM/DDIM): **20× training cost**, yet fails to converge.
- **DiffResist**: best on **all metrics**, **smallest model** (6.71M), **lowest FLOPs**.
- Inference: **0.44 s** vs. 6.45 s per tile $\Rightarrow >10\times$ **faster**.

Qualitative Results

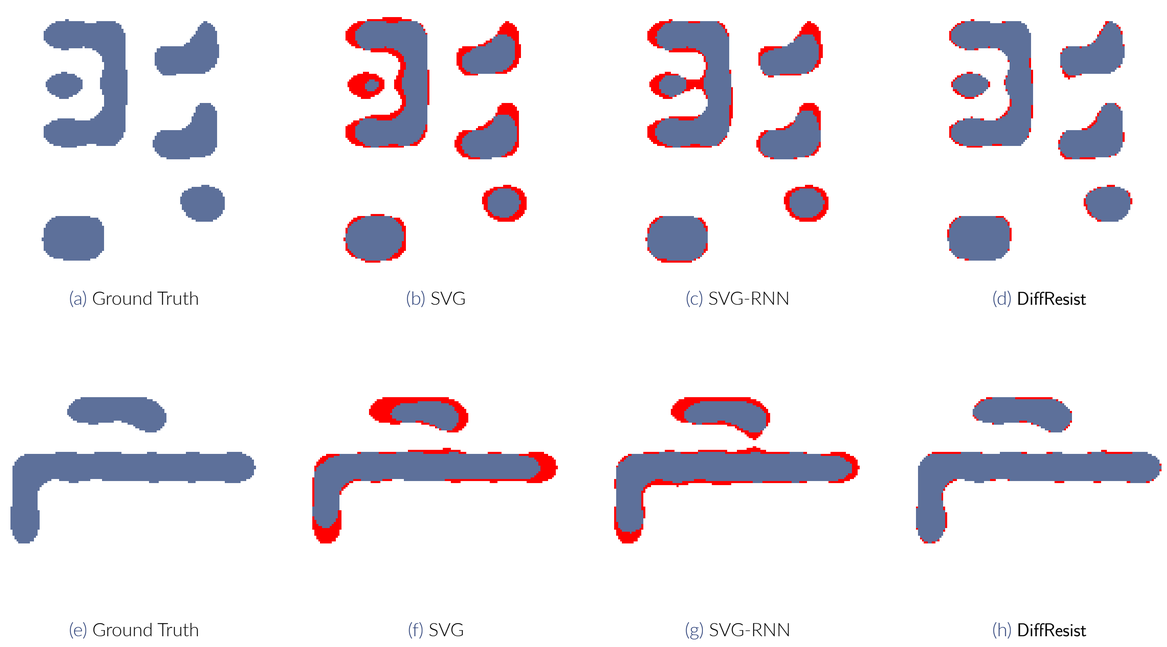

Figure 3. Predicted printed images (difference from ground truth highlighted). **DiffResist** produces the closest match.

Ablation Study & Error Propagation

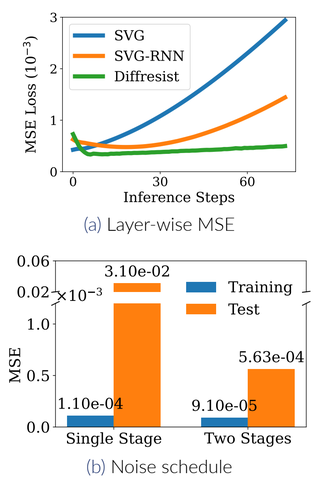

Figure 4. Ablation: (a) Layer-wise MSE — baseline errors grow rapidly; **DiffResist** stays stable via boundary condition. (b) Two-stage vs. linear noise schedule — similar training loss, but two-stage yields lower test MSE.

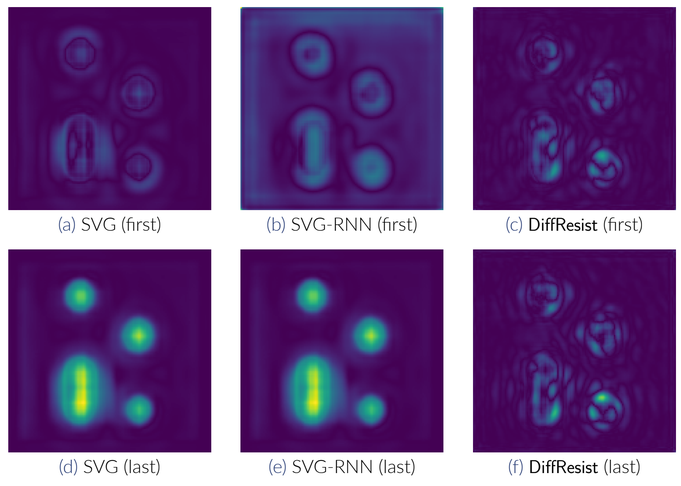

Figure 5. Error heatmaps at the **first** (top) / **last** (bottom) inference step. Baseline errors amplify across depth; **DiffResist** stays bounded by the physical boundary condition.

Conclusion

DiffResist reformulates 3D resist prediction as a physics-constrained 2D diffusion problem. By aligning diffusion steps with physical depth, injecting the resist–air boundary condition, and confining stochasticity to a warm-up segment, it mitigates error accumulation. With lightweight super-resolution, it achieves **state-of-the-art accuracy** and $>10\times$ **faster inference** on LithoBench.

References

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